E Appendix E: Ablation Studies

In order to validate our results, we conducted multiple different trials with our flagging algorithm, specifically the cap of our false positive rate, to analyze the performance of our model with different hyper parameters.

Table 5: ZEDD Results for each model with **clean false positive cap at 5%**. Values shown as % flagged.

Model	C	EM	J	PC	SL	TO
Sentence BERT (All-MPNET-BASE-V2)	2.2%	95.9%	86.2%	90.5%	91.6%	86.8%
Llama 3 8B Instruct	5.4%	98.1%	92.2%	94.2%	96.8%	91.0%
Mistral 7B Instruct	3.4%	98.2%	92.2%	93.3%	96.9%	90.9%
Qwen 2 7B Instruct	5.4%	98.2%	91.7%	94.1%	96.9%	90.4%

Table 6: ZEDD Results for each model with **clean false positive cap at 10%**. Values shown as % flagged.

Model	C	EM	J	PC	SL	TO
Sentence BERT (All-MPNET-BASE-V2)	8.1%	96.0%	86.2%	90.6%	91.6%	86.8%
Llama 3 8B Instruct	5.4%	98.1%	92.2%	94.2%	96.8%	91.0%
Mistral 7B Instruct	5.4%	98.2%	92.2%	93.3%	96.9%	90.9%
Qwen 2 7B Instruct	5.4%	98.2%	91.7%	94.1%	96.9%	90.4%

Observations: Evident from our ablation studies, the training of the Gaussian Mixture Model (GMM) is more effective at lower thresholds in comparison with higher thresholds as it significantly reduced the false positives reported by the GMM. Between the 5% threshold and the 10% threshold, the GMM performed as expected, increasing the overall flag rate and thus flagging more prompt pairs that are on the lower end of the tail in the distribution of embeddings, evident by the larger False Positive Rate (C%) in the 10% false positive cap.

F Appendix F: Dataset Creation and Preparation

F.1 Prompt Pair Generation

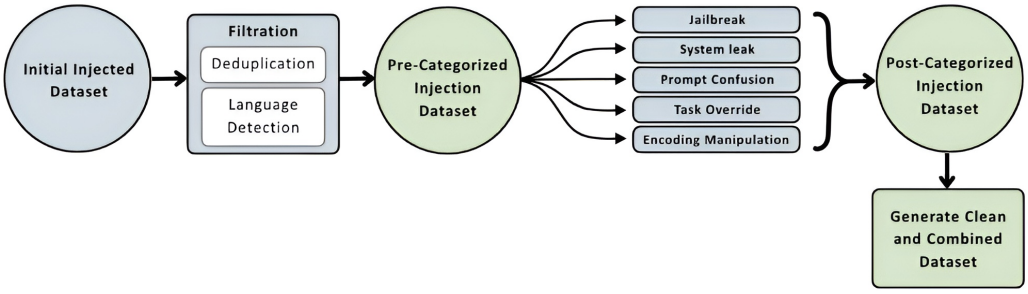

Figure 8: ZEDD Data Processing Pipeline

We use the Microsoft LLMail-Inject Dataset [25], which contains adversarial emails targeting LLM-integrated assistants via indirect prompt injection. To support drift analysis later in our pipeline, we generate a dataset of aligned adversarial-clean prompt pairs, applying the following preprocessing pipeline:

Deduplication and Language Filtering. We deduplicate the data and filter out prompts in any language other than English with FastText's *lid.176.ftz* language identification model [26, 27]. We only keep the unique English prompts that contain the term **"system"** (capturing system prompt leakage attempts).

F.1.1 Injection Classification

For stratified evaluation, we use GPT-3.5-turbo-0125 to label each prompt as one of "jailbreak", "system leak", "task override", "encoding manipulation", and "prompt confusion."

By creating category classifications, we make the ZEDD technology adaptable to different scenarios depending on the type of injection.

F.1.2 Clean Prompt Generation

Each filtered injected prompt is paired with a clean variant using a constrained LLM-based rewrite. We employ a custom writing function that utilizes the OpenAI Batch API to create calls to the *GPT-3.5-turbo-0125* model, similar to section F.1.1, with a system-level safety prompt aimed at preserving the original task semantics while eliminating malicious or override behavior. This results in aligned injected and clean prompt pairs, suitable for drift analysis.

F.1.3 Dataset Reduction and Fully Clean Prompt Pair Generation

We subsample **around 86,000** injected–clean pairs and generate an additional **86,000** clean–clean pairs using the *OpenAI Batch API* to provide a baseline for embedding calculations. The unused portion of the dataset is reserved for evaluation. **Clean–clean** pairs are labeled with the category *"clean"* to distinguish them from **injected–clean** pairs.

For training the ZEDD embedding model, we assign *similar* to **clean–clean** pairs and *not similar* to **injected–clean** pairs. These labels serve as ground-truth labels for semantic similarity detection.

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